Andrew Jacobson

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EDUCATION

University of Massachusetts Lowell, Lowell, MA — September 2021 - May 2025
Bachelor of Science: Computer Science, Mathematics Minor

TECHNICAL SKILLS

- **Programming:** C/C++, Python, Go, SQL, Bash, Data Structures & Algorithms
- **Backend:** FastAPI, Rest APIs, Paramiko, AsyncIO/asyncpg, SQLAlchemy, PostgreSQL, Redis
- **Infrastructure:** Linux networking & Administration, Git, Docker / Docker Compose, GDB / Valgrind, Yocto Project, Zabbix, Grafana
- **Networking:** TCP/IP, SDN/SD-WAN, OSPF, EVPN, FRR, L2TP, Network Namespaces, Policy Routing, Netfilter, DHCP/DNS, eBPF/XDP

EXPERIENCE

WiLine Networks, Remote
Software / Operations Engineer, SDWAN Platform — June 2025 - August 2026

- Serve as a primary operator of WiLine's proprietary SD-WAN controller platforms, configuring standard and custom device/service profiles and maintaining secure overlay connectivity for thousands of customer SD-WAN devices and aggregation gateways across mixed wired and wireless transports.
- Built the internal FastAPI platform (Python, PostgreSQL, Redis) unifying fleet monitoring, provisioning, and diagnostics across the WiLine Network Controller fleet and replacing manual operations workflows; built the async data layer (SQLAlchemy, asyncpg) and in-process workers for uptime tracking and caching.
- Built an SSH-based fleet automation engine (Paramiko) running versioned playbooks through a jump host with Server Sent Event progress streaming; built for automated diagnostics, health checks, and remediations.
- Maintained Zabbix monitoring across the SD-WAN fleet and authored custom alerting templates for customers, cutting alert noise and speeding incident detection.
- Triaged fleet and network incidents across WiLine Network Controllers, aggregation-gateway routing (FRR, OSPF, EVPN), and CPE L2TP tunnels, correlating WNC agent logs/diagnostics with telemetry to drive root cause to remediation.
- Shipped Python, Go, and Bash automation and internal tooling that replaced manual runbooks for diagnostics and firmware/software rollouts.

Network/Systems Engineer, Intern — June 2024 - June 2025

- Wrote Python and Bash automation to discover and normalize UPS configuration for Zabbix across a large device dataset; auto-onboarded 517 of 567 devices and flagged unreachable or misconfigured units for remediation.
- Ran root-cause diagnostics and validation testing on customer SD-WAN edge devices over LTE/5G and wired backhaul, isolating issues with Linux networking tools and Zabbix telemetry before handing off to NOC.
- Analyzed signal and throughput data across ~300 LTE/5G sites and produced data-driven upgrade recommendations and reporting for leadership.

PROJECTS

encvr

- Encrypted point-to-point communication system on ESP32 using LoRa radio modules, with firmware structured as FreeRTOS tasks and a custom AT-command protocol layer over UART for module configuration, message TX/RX, and signal-quality reporting.
- Implemented interrupt-driven GPIO input, a hardware-timer module for radio control signaling, an I2C OLED status display, and ADC-sampled input for live on-device reconfiguration without reflashing.

tunnelMTU

- Statically linked Go CLI for embedded Yocto CPE hardware that detects MTU fragmentation in L2TP/5G backhaul using binary-search ICMP path-MTU discovery against live tunnels; sets up network namespaces, veth pairs, per-tunnel policy routing, and SNAT.

myFridge

- SwiftUI iOS app that tracks food by expiration date, parsing grocery receipts on-device with Vision OCR and Apple Intelligence.

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- Ansible-managed homelab: OpenWrt router with VLAN firewall zones, 3-node Proxmox VE cluster on ZFS, Tailscale, and Zabbix.
- Self-hosted website on a Proxmox LXC behind Cloudflare Tunnel, built with Caddy, a Node.js API, and GitHub Actions CI/CD.

flowView

- XDP program in C that classifies TCP/UDP flows and redirects them into an AF_XDP socket via XSKMAP. Go userspace service (cilium / eBPF) attaches the program, manages the fill/RX rings, inspects packets, and aggregates live per-flow statistics.

fieldNotes

- Early-stage wearable push-to-talk note device on ESP32 with on-device open-source speech-to-text for timestamped notes.